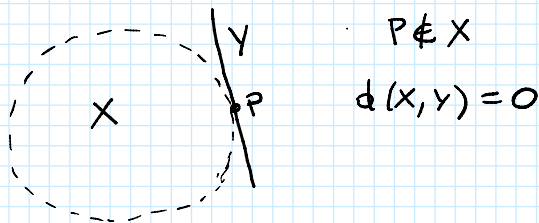


$$(\vec{E}, \mathcal{E}, \pi) \quad \dim \mathcal{E} = m \quad \mathcal{R} = (O, \mathcal{B})$$

$$P, Q \in \mathcal{E}, \quad d(P, Q) \stackrel{\text{def}}{=} \|\overrightarrow{PQ}\|$$

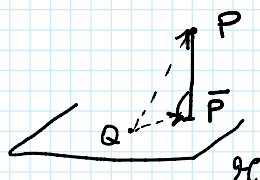
$$X, Y \subseteq \mathcal{E}, \quad d(X, Y) \stackrel{\text{def}}{=} \inf \{d(P, Q) \mid P \in X, Q \in Y\}$$



Siano $P \in \mathcal{E}$ e $\mathcal{H}$ un iperpiano

Se $P \in \mathcal{H}$, allora $d(P, \mathcal{H}) = d(P, P) = 0$

Se $P \notin \mathcal{H}$, allora:



$$d(P, \mathcal{H}) = d(P, \bar{P})$$

Esempio: $m = 4$

$$\mathcal{R} = (O, \mathcal{B})$$

$$\mathcal{H}: x_1 - 3x_2 - x_3 + 2x_4 - 1 = 0$$

$$w(1, -3, -1, 2) \perp \mathcal{H}, \quad \perp \mathcal{H} = \mathcal{L}(w)$$

$$P(0, 1, 2, 0)$$

$$\mathcal{R}: \begin{cases} x_1 = t \\ x_2 = 1 - 3t \\ x_3 = 2 - t \\ x_4 = 2t \end{cases} \quad T\left(t, 1 - 3t, 2 - t, 2t\right)$$

$$\mathcal{R} \cap \mathcal{H}: t - 3 + 9t - 2 + t + 4t - 1 = 0$$

$$15t - 6 = 0 \Rightarrow \bar{t} = \frac{6}{15} = \frac{2}{5}$$

$$\bar{P}\left(\frac{2}{5}, -\frac{1}{5}, \frac{8}{5}, \frac{4}{5}\right)$$

$$d(P, \mathcal{H}) = d(P, \bar{P}) = \|\overrightarrow{P\bar{P}}\| = \sqrt{\frac{4}{25} + \frac{36}{25} + \frac{64}{25} + \frac{16}{25}} = \sqrt{\frac{60}{25}} = \frac{\sqrt{60}}{5}$$

$$\overrightarrow{P\bar{P}}\left(\frac{2}{5}, -\frac{6}{5}, -\frac{2}{5}, \frac{4}{5}\right)$$

In generale:

$$\mathcal{H}: a_1x_1 + a_2x_2 + \dots + a_nx_n - b = 0$$

$$P(c_1, c_2, \dots, c_n)$$

$$(a_1, a_2, \dots, a_n) \neq 0$$

$$w(a_1, a_2, \dots, a_n) \quad \perp \mathcal{H} = \mathcal{L}(w)$$

$$\mathcal{R}: \begin{cases} x_1 = c_1 + a_1t \\ x_2 = c_2 + a_2t \\ \vdots \\ x_n = c_n + a_nt \end{cases}$$

$$\mathcal{R} \perp \mathcal{H} \quad \text{e} \quad P \in \mathcal{R}$$

$$T(c_1 + a_1 t, c_2 + a_2 t, \dots, c_n + a_n t)$$

$$a_1(c_1 + a_1 t) + a_2(c_2 + a_2 t) + \dots + a_n(c_n + a_n t) - b = 0$$

$$a_1 c_1 + a_1^2 t + a_2 c_2 + a_2^2 t + \dots + a_n c_n + a_n^2 t - b = 0$$

$$(a_1^2 + a_2^2 + \dots + a_n^2)t = b - a_1 c_1 - a_2 c_2 - \dots - a_n c_n$$

$$\bar{t} = \frac{b - a_1 c_1 - a_2 c_2 - \dots - a_n c_n}{a_1^2 + a_2^2 + \dots + a_n^2}$$

$$\bar{P}(c_1 + a_1 \bar{t}, c_2 + a_2 \bar{t}, \dots, c_n + a_n \bar{t})$$

$$\overrightarrow{P\bar{P}}(a_1 \bar{t}, a_2 \bar{t}, \dots, a_n \bar{t})$$

$$\begin{aligned} \|\overrightarrow{P\bar{P}}\| &= \sqrt{a_1^2 \bar{t}^2 + a_2^2 \bar{t}^2 + \dots + a_n^2 \bar{t}^2} = \sqrt{(a_1^2 + a_2^2 + \dots + a_n^2) \frac{(b - a_1 c_1 - a_2 c_2 - \dots - a_n c_n)^2}{(a_1^2 + a_2^2 + \dots + a_n^2)^2}} = \\ &= \frac{|a_1 c_1 + a_2 c_2 + \dots + a_n c_n - b|}{\sqrt{a_1^2 + a_2^2 + \dots + a_n^2}} = d(P, \bar{P}) = d(P, \mathcal{H}) \end{aligned}$$

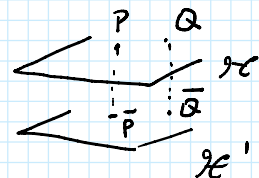
Esempio $m=2$ $R=(0, 0)$

$$\mathcal{H}: 3x_1 + 2x_2 - 4 = 0 \quad P(1, -2)$$

$$d(P, \mathcal{H}) = \frac{|3-4-4|}{\sqrt{9+4}} = \frac{5}{\sqrt{13}}$$

$\dim \mathcal{E} = m$ $R=(0, 0)$

$\mathcal{H}, \mathcal{H}'$ iperpiani paralleli: $\mathcal{H} = \mathcal{H}'$



$$\mathcal{H}: a_1 x_1 + a_2 x_2 + \dots + a_n x_n - b = 0$$

$$\mathcal{H}': a_1 x_1 + a_2 x_2 + \dots + a_n x_n - b' = 0$$

?

$$\|\overrightarrow{P\bar{P}}\| = \|\overrightarrow{Q\bar{Q}}\|$$

$$P(c_1, c_2, \dots, c_n) \in \mathcal{H}$$

$$Q(d_1, d_2, \dots, d_n) \in \mathcal{H}'$$

$$d(P, \mathcal{H}') = \frac{|a_1 c_1 + a_2 c_2 + \dots + a_n c_n - b'|}{\sqrt{a_1^2 + \dots + a_n^2}} = \frac{|b - b'|}{\sqrt{a_1^2 + \dots + a_n^2}}$$

$$P \in \mathcal{H} \Rightarrow a_1 c_1 + a_2 c_2 + \dots + a_n c_n = b$$

$$d(Q, \mathcal{H}') = \frac{|a_1 d_1 + \dots + a_n d_n - b'|}{\sqrt{a_1^2 + \dots + a_n^2}} = \frac{|b - b'|}{\sqrt{a_1^2 + \dots + a_n^2}}$$

$$\begin{aligned} d(\mathcal{H}, \mathcal{H}') &= d(P, \mathcal{H}') \quad \forall P \in \mathcal{H} \\ &= d(\mathcal{H}, R) \quad \forall R \in \mathcal{H}' \end{aligned}$$

$$d(\mathcal{H}, \mathcal{H}') = d(P, \mathcal{H}') \quad \forall P \in \mathcal{H}$$

$$\stackrel{||}{=} d(\mathcal{H}, R) \quad \forall R \in \mathcal{H}'$$

Esempio: $n=2$

$$\mathcal{L}: -x_1 + 3x_2 - 2 = 0$$

$$\mathcal{L}': 2x_1 - 6x_2 + 3 = 0$$

$$\vec{\mathcal{L}}: -x_1 + 3x_2 = 0$$

$$\vec{\mathcal{L}}': 2x_1 - 6x_2 = 0$$

$$\mathcal{L}': -x_1 + 3x_2 - \frac{3}{2} = 0$$

$$\vec{\mathcal{L}} = \vec{\mathcal{L}}' \Rightarrow \mathcal{L} // \mathcal{L}'$$

$$P(1,1) \in \mathcal{L}$$

$$d(\mathcal{L}, \mathcal{L}') = \frac{|-2 + \frac{3}{2}|}{\sqrt{1+9}} = \frac{\frac{1}{2}}{\sqrt{10}} = \frac{1}{2\sqrt{10}}$$

$$\stackrel{||}{=} \frac{|2-6+3|}{\sqrt{4+36}} = \frac{1}{\sqrt{40}} = \frac{1}{2\sqrt{10}}$$

Esempio $\mathcal{E} = \mathbb{R}^2$ $\vec{\mathcal{E}} = \mathbb{R}^2$ col prodotto scalare standard

$$\mathcal{R} = (O = (2, -3), \mathcal{B} = ((\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}), (\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})))$$

$$\mu = (0,0) \quad \mu \equiv_{\mathcal{R}} \phi_{\mathcal{B}}(\overrightarrow{\mu O}) = \phi_{\mathcal{B}}(\underline{(2, -3)})$$

$(\vec{\mathcal{E}}, \mathcal{E}, \pi)$ $\dim \mathcal{E} = 3$ $\mathcal{R} = (O, \mathcal{B})$

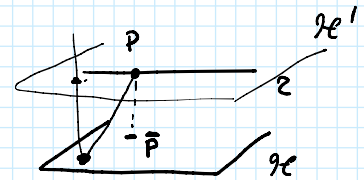
$$\mathcal{H}: a_1x_1 + a_2x_2 + a_3x_3 - b = 0$$

$$\mathcal{L}: \begin{cases} a_1x_1 + a_2x_2 + a_3x_3 - \beta = 0 \\ a'_1x_1 + a'_2x_2 + a'_3x_3 - \beta' = 0 \end{cases}$$

$$\mathcal{L} \cap \mathcal{H} \neq \emptyset, \text{ allora } d(\mathcal{L}, \mathcal{H}) = 0$$

Altrimenti: $\mathcal{L} // \mathcal{H}$

$$\forall P \in \mathcal{L}, \quad d(P, \mathcal{H}) = d(\mathcal{L}, \mathcal{H})$$



Esercizio: $\mathcal{R} = (O, \mathcal{B})$ $\mathcal{H}: 2x_1 - x_2 + 2x_3 - 1 = 0$

(i) determina una retta $\mathcal{L}$: $\mathcal{L} // \mathcal{H}$ e $P(0,2,1) \in \mathcal{L}$ ($P \notin \mathcal{H}$)

(ii) determina la distanza tra $\mathcal{L}$ e $\mathcal{H}$.

(i) $\vec{\mathcal{H}}: 2x_1 - x_2 + 2x_3 = 0$ $\mu(1,0,-1)$

$$\mathcal{L}: \begin{cases} x_1 = t \\ x_2 = 2 \\ x_3 = 1 - t \end{cases}$$

(ii) $d(\mathcal{L}, \mathcal{H}) = d(P, \mathcal{H}) = \frac{|-2+2-1|}{\sqrt{4+1+4}} = \frac{1}{3}$

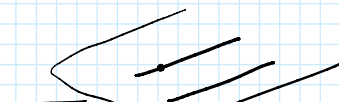
$(\vec{\mathcal{E}}, \mathcal{E}, \pi)$ $\dim \mathcal{E} = 3$ $\mathcal{R} = (O, \mathcal{B})$

$\mathcal{L}, \mathcal{L}'$ due rette

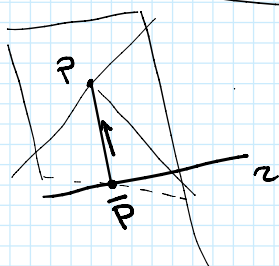
• $\mathcal{L} \cap \mathcal{L}' \neq \emptyset \Rightarrow d(\mathcal{L}, \mathcal{L}') = 0$

• $\mathcal{L} // \mathcal{L}' \Rightarrow d(\mathcal{L}, \mathcal{L}') = d(P, \mathcal{L}') \quad \forall P \in \mathcal{L}$

$$\dim^\perp \vec{\mathcal{L}} = 2$$



• $r // r' \Rightarrow d(r, r') = d(P, r') \quad \forall P \in r$



$\mathcal{H}$ piano ortogonale a r e passante per P
 $\bar{P} = \mathcal{H} \cap r$
 $d(P, r) = d(P, \bar{P})$

Esempio:

$m=3$

$r: \begin{cases} x_1 = 3 + t \\ x_2 = -2 - t \\ x_3 = 1 + 2t \end{cases}$

$P(0, 0, 1) \notin r$
 $u(1, -1, 2)$

$r': r' \perp r \text{ e } |r' \cap r| = 1$



$\vec{r}' \subseteq \perp \vec{r} \Rightarrow r' \subseteq \text{piano ortogonale a } r \text{ e passante per } P$

$\mathcal{H}: x_1 - x_2 + 2x_3 + k = 0$

$\mathcal{H}: x_1 - x_2 + 2x_3 - 2 = 0$

$0 - 0 + 2 + k = 0 \Rightarrow k = -2$

$\bar{P} = r \cap \mathcal{H}$

$T(3+t, -2-t, 1+2t) \in \mathcal{H} \Rightarrow$

$\Rightarrow 3+t-2-t+2(1+2t)-2=0 \Rightarrow 6t+5=0 \Rightarrow t = -5/6$

$\bar{P}(3-5/6, -2+5/6, 1-10/6)$

$d(P, r) = d(P, \bar{P})$

• r e r' sghembe

Teorema della comune perpendicolare:

r e r' rette sghembe. Esiste un'unica retta s che è ortogonale sia a r che a r' ed è incidente sia a r che a r' :

$\exists!$ retta $s: s \perp r \text{ e } s \perp r'$

$P = s \cap r \neq \emptyset \text{ e } P' = s \cap r' \neq \emptyset$

Inoltre:

$d(r, r') = d(P, P')$

DIM

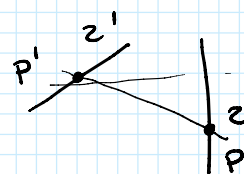
$B = (0, B)$ B ortonormale

$r: \begin{cases} x = x_0 + lt \\ y = y_0 + mt \\ z = z_0 + nt \end{cases}$

$u(l, m, n)$

$r': \begin{cases} x = x'_0 + l't' \\ y = y'_0 + m't' \\ z = z'_0 + n't' \end{cases}$

$u'(l', m', n')$



$P(x_0 + lt, y_0 + mt, z_0 + nt) \in r$

$P'(x'_0 + l't', y'_0 + m't', z'_0 + n't') \in r'$

Imponiamo che: $\overrightarrow{PP'} \perp r \Leftrightarrow \langle \overrightarrow{PP'}, u \rangle = 0$
 e $\overrightarrow{PP'} \perp r' \Leftrightarrow \langle \overrightarrow{PP'}, u' \rangle = 0$

(*)
 $\Leftrightarrow$

$\vec{u} \perp \vec{u}'$

$$e \quad PP' \perp z' \Leftrightarrow \langle PP', u' \rangle = 0$$

$$\overrightarrow{PP'}(x_0' + l'z' - x_0 - lt, y_0' + m'z' - y_0 - mt, z_0' + n'z' - z_0 - nt)$$

$$(*) \Leftrightarrow \begin{cases} (x_0' + l'z' - x_0 - lt)l + (y_0' + m'z' - y_0 - mt)m + (z_0' + n'z' - z_0 - nt)n = 0 \\ (x_0' + l'z' - x_0 - lt)l' + (y_0' + m'z' - y_0 - mt)m' + (z_0' + n'z' - z_0 - nt)n' = 0 \end{cases}$$

$$\Leftrightarrow \begin{cases} (l^2 + m^2 + n^2)z' + (-l^2 - m^2 - n^2)z + \dots = 0 \\ (l'l' + m'm + n'n)z' + (-l'l' - m'm' - n'n')z + \dots = 0 \end{cases}$$

$\|u'\|^2 \qquad \qquad \qquad -\langle u, u' \rangle$

$$\det \begin{pmatrix} \langle u, u' \rangle & -\|u\|^2 \\ \|u'\|^2 & -\langle u, u' \rangle \end{pmatrix} = -\langle u, u' \rangle^2 + \|u\|^2 \|u'\|^2$$

se $\det(A) \neq 0$, per il Teorema di Cramer $\exists! (t', t)$ che soddisfa il sistema,
perciò $\exists! (P', P)$ che soddisfa il Teorema

se $\det(A) = 0$, allora $\langle u, u' \rangle^2 = \|u\|^2 \|u'\|^2$, ossia $|\langle u, u' \rangle| = \|u\| \|u'\|$

ossia la disuguaglianza di Schwarz vale come uguaglianza

e questo è possibile se e solo se $\{u, u'\}$ è lin. dipendente: $u \| u' \leadsto$
ASSURDO.